

YASH KARTHIYA

Backend Developer | B.Tech Student

Phone: +91 87990 06562 | Email: yash.karthiya.1066@gmail.com | Location: Gujarat, India

LinkedIn: linkedin.com/in/yash-karthiya | GitHub: github.com/NotYash1066 | Portfolio: yashkarthiya.in

Computer Engineering student building Java/Spring and Node/FastAPI backends, with project experience in multi-tenant architectures, asynchronous processing, WebSockets, and retrieval systems.

ENGINEERING HIGHLIGHTS

- Designed **high-throughput APIs** load-tested for 500+ concurrent users with sub-50ms response times.
- Reduced average query latency from 150ms to **<50ms** by implementing Redis L1/L2 caching.
- Built **event-driven architectures** leveraging RabbitMQ to decouple services and improve fault tolerance.
- Developed **multi-tenant systems** using Row-Level Security (RLS) to ensure strict data isolation.

TECHNICAL EXPERTISE

Languages	Java (Core & Spring), Python, JavaScript (Node.js), SQL (PostgreSQL)
Backend	Spring Boot, Spring Cloud, Hibernate, Express, FastAPI, JWT Auth, REST APIs
Databases	PostgreSQL (Schema Design/Indexing), MongoDB, Redis (Advanced Caching)
Infra & Testing	AWS (EC2, S3), Docker, RabbitMQ, Linux, JUnit, Mockito, Git, CI/CD Basics

EDUCATION

Bachelor of Technology in Computer Engineering	2023 – 2027
Dharmsinh Desai University CGPA: 8.29 / 10.0	Nadiad, Gujarat

KEY PROJECTS

Scalable Multi-Tenant Inventory System	Dec 2024 – Mar 2025
<i>Spring Boot, PostgreSQL, Redis, Java 17, Docker, JUnit, Mockito</i> • Designed a multi-tenant architecture for shop management using PostgreSQL schemas and Row-Level Security (RLS) to ensure zero cross-tenant data leakage. • Achieved <50ms response times for high-traffic SKUs by implementing a write-behind Redis caching strategy, load-tested using Locust to simulate peak seasonal traffic. • Ensured 90% code coverage for critical service layers by writing unit tests with JUnit and Mockito; containerized the entire stack for consistent deployment on AWS EC2.	View on GitHub
SkillSwap: Real-Time Signaling Backend	Jul 2025 – Nov 2025
<i>Node.js, Express, MongoDB, Socket.io, WebRTC, JWT Auth</i> • Engineered a real-time signaling server for P2P video sessions, handling 100+ concurrent WebSocket connections via an event-driven architecture. • Implemented secure authentication using JWT with role-based access control (RBAC) and input validation to prevent unauthorized session entry. • Added fault-tolerance through a heartbeat-based cleanup mechanism that purges stale session data and notifies peers within 30 seconds of ungraceful disconnects.	View on GitHub
RAG-Based Academic Query Engine	Dec 2025 – Mar 2026
<i>FastAPI, Python, ChromaDB, LangChain</i> • Optimized retrieval latency by 30% across 500+ documents using HNSW indexing and recursive character splitting for semantic search queries. • Built a high-throughput retrieval pipeline achieving sub-500ms response times by fine-tuning chunk sizes and similarity search parameters.	View on GitHub

HACKATHONS & EVENTS

DUHacks 5.0 Top 10 Finalist (100+ Teams)	2026
<i>Food Label AI Scanner</i> • Built an automated OCR-to-NLP pipeline to classify food ingredients from packaging; implemented a weighted health-scoring algorithm validated on 200+ samples.	Tech: FastAPI, Tesseract, OpenAI
Holboxai Hackathon Top 5 Team	2025
<i>NeuroHire</i> • Developed a resume screening pipeline using LangChain to extract unstructured text into profiles, significantly improving shortlisting efficiency during hackathon testing.	Tech: Python, OCR, LangChain, LLMs

LANGUAGES

English (Professional) • Hindi (Native) • Gujarati (Native)